

Intent Detection Evaluation Report

gemini-3.5-flash on the AfroBench-Lite intent detection task

Generated from the summary results and the detailed instance-level predictions in the experiment directory.

Abstract

This report summarizes a zero-shot intent-detection evaluation of gemini-3.5-flash across 14 AfroBench-Lite languages. The aggregate metrics come from intent_detection_results.csv, and the supporting instance-level evidence comes from intent_detection_combined_results.csv.

Methodology

The experiment evaluated five prompt templates over 14 languages. Performance is reported using both F1 and accuracy. The tables below show the best-performing prompt per language, the full prompt-by-language matrix, and the macro-average F1/accuracy by prompt.

Table 1: Best prompt per language

Language	Best Prompt	Best F1	Best Acc
amh	prompt_3	0.9297	0.9297
eng	prompt_5	0.8875	0.8875
hau	prompt_1	0.9625	0.9625
ibo	prompt_5	0.9313	0.9313
kin	prompt_4	0.8266	0.8266
lin	prompt_4	0.9266	0.9266
lug	prompt_4	0.9031	0.9031
orm	prompt_2	0.8719	0.8719
sna	prompt_2	0.9609	0.9609
sot	prompt_5	0.7922	0.7922
swa	prompt_5	0.9359	0.9359
xho	prompt_1	0.9062	0.9062
yor	prompt_2	0.8953	0.8953
zul	prompt_3	0.8656	0.8656

Table 2: Prompt-by-language performance matrix

Language	prompt_1	prompt_2	prompt_3	prompt_4	prompt_5	Best Prompt	Best F1
amh	0.9234	0.9281	0.9297	0.9266	0.9266	prompt_3	0.9297
eng	0.8842	0.8778	0.8778	0.8778	0.8875	prompt_5	0.8875
hau	0.9625	0.9625	0.9625	0.9594	0.9609	prompt_1	0.9625
ibo	0.9250	0.9266	0.9234	0.9281	0.9313	prompt_5	0.9313
kin	0.8250	0.8250	0.8219	0.8266	0.8266	prompt_4	0.8266
lin	0.9156	0.9156	0.9125	0.9266	0.9219	prompt_4	0.9266
lug	0.8938	0.8797	0.8859	0.9031	0.8938	prompt_4	0.9031

Language	prompt_1	prompt_2	prompt_3	prompt_4	prompt_5	Best Prompt	Best F1
orm	0.8625	0.8719	0.8656	0.8703	0.8656	prompt_2	0.8719
sna	0.9547	0.9609	0.9609	0.9594	0.9578	prompt_2	0.9609
sot	0.7812	0.7844	0.7859	0.7844	0.7922	prompt_5	0.7922
swa	0.9297	0.9250	0.9313	0.9250	0.9359	prompt_5	0.9359
xho	0.9062	0.9016	0.9047	0.9016	0.9062	prompt_1	0.9062
yor	0.8906	0.8953	0.8859	0.8875	0.8938	prompt_2	0.8953
zul	0.8641	0.8625	0.8656	0.8594	0.8547	prompt_3	0.8656

Table 3: Macro-average F1 and accuracy by prompt

Prompt	Macro-average F1	Macro-average Acc
prompt_1	0.8942	0.8942
prompt_2	0.8941	0.8941
prompt_3	0.8938	0.8938
prompt_4	0.8954	0.8954
prompt_5	0.8968	0.8968

Discussion

Prompt_5 is the strongest single fixed prompt overall, while the best prompt per language varies by language. This indicates that prompt choice remains meaningful even when the aggregate performance is high. The detailed sample-level evidence in `intent_detection_combined_results.csv` complements the summary table and provides the underlying per-example predictions.

Evidence files

The experiment artifacts are stored in the same folder as the report: `intent_detection_results.csv` contains the summary metrics, `intent_detection_combined_results.csv` contains the detailed predictions, and this PDF consolidates both into a readable report.